Saleep Shrestha

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Education

The University of Southern Mississippi (USM), Hattiesburg, MS May 2027
Bachelor of Science in Computer Science, Honors Discovery Scholar, IBM Z Ambassador, 5x President's List GPA: 3.971
Relevant Courses: Data Structures and Algorithms, Operating Systems, Database Management

Experience

Software Engineer (Client-Facing) Aug 2025 - Present
illumibot Hattiesburg, MS

- Built an AI support workflow using Claude and the Metricool API that answers posts and DMs from a knowledge base 3 times a day, cutting in-app tickets 50% from Halloween week 2025 to 2026 despite 2-3x user growth
- Owned the illumiBeast launch, building the website, forms, and automatic lead emails using React, TypeScript, and Supabase for a 31-city, 31-day tour that generated 73 sales leads and 5 paid service requests
- Designed a QR attribution system with PostgreSQL and Edge Functions that ties each scan to one of 5 trucks and logs location and call or email clicks, reported through an admin dashboard
- Built a projector recommendation engine that matches users to models in our database using house size, throw distance, and ambient light, increasing partner projector (BenQ) sales by 15%
- Shipped Flutter and React features across Gallery, Timeline, and AI Corner Pinning for an app with 11,800+ users and 2,500+ active subscribers, leading a team of 5 through task assignment, code reviews, and beta releases

Undergraduate Researcher (Honors Thesis) Apr 2026 - Present
Graduate Admissions AI Assistant, School of Computing Sciences and Engineering, USM Hattiesburg, MS

- Built a Corrective-RAG agent using Python, LangGraph, FastAPI, and Qdrant that answers admissions questions from official sources, validates each claim against retrieved text, and refers out what it can't verify
- Implemented hybrid dense and sparse retrieval with cross-encoder reranking, plus a semantic answer cache that returns hits in under a second versus a 12.7 s mean for the full pipeline
- Scored 21 of 25 fully correct with no confidently wrong answers on an adversarial test set, with 142 automated tests and CI

Software Development Associate Apr 2025 - May 2026
ArroyoDev Hattiesburg, MS

- Revamped the Retina MS and Eyecare Associates websites in Vue and TypeScript, adding insurance verification features
- Engineered patient intake forms handling 1,000+ patients annually in a HIPAA-regulated clinic environment
- Built and maintained backend data workflows for patient intake, storage, and clinic operations, shipping production updates from requirements gathered with clinical staff

Projects

MrGrammar Mar 2025 - Present
Creator, Chrome Extension Chrome Web Store

- Shipped a Manifest V3 Chrome extension that corrects selected text in place across Gmail, Outlook, and Slack
- Designed a serverless proxy on Vercel that keeps API keys off the client, calls Gemini with automatic Groq fallback, and enforces rate limits with Upstash Redis
- Hardened security by rejecting unknown origins, failing closed, and inserting model output only as text, with automated tests

US Marine Corps (Hacking for Defense) Jan 2025 - May 2025
Team Lead, Data Engineer

- Conducted 90+ direct stakeholder interviews across the USMC, US Army, the Pentagon, Esri, and Palantir to identify workflow bottlenecks and translate requirements into an MVP
- Engineered a Python ETL geospatial pipeline using Census APIs (TIGER shapefiles), Bing Maps, and ArcGIS to geocode 30,000+ recruit addresses and align them with congressional districts
- Automated district-level recruiting reports for Congress and delivered Power BI dashboards used by the Marine Corps

Awards & Leadership

Social Impact Winner, Hatchathon powered by Voxo 2025 | IBM Z Ambassador: organized a campus-wide IBM Z career event | IBM Z Xplore Concepts Certificate

Technical Skills

Languages: Python, C++, TypeScript, JavaScript, Dart, SQL, COBOL
Frontend & Mobile: React, Vue, Flutter
Backend, API & Database: Node.js, FastAPI, REST APIs, Supabase, PostgreSQL, MySQL, Redis, Qdrant
AI & Data: RAG, LangGraph, OpenAI API, Claude, NumPy, ETL, Power BI, ArcGIS
Cloud, Systems & Tools: Vercel, Firebase, Docker, Linux, Git, GitHub Actions, pytest, IBM z/OS, JCL